

```
[**] [1:2010935:2] ET POLICY Suspicious inbound
to MSSQL port 1433 [**]
[Classification: Potentially Bad Traffic]
[Priority: 2]
07/25-20:31:31.817217 232.11.237.105:1000 ->
233.29.20.24:1433
TCP TTL:102 TOS:0x0 ID:256 IpLen:20 DgmLen:40
*****S* Seq: 0x43EE0000 Ack: 0x0 Win: 0x4000
TcpLen: 20
[Xref => http://doc.emergingthreats.net/2010935]
```

```
[**] [1:2010935:2] ET POLICY Suspicious inbound
to MSSQL port 1433 [**]
[Classification: Potentially Bad Traffic]
[Priority: 2]
07/25-20:31:31.838622 17.1.27.05:6000 ->
23.12.20.17:1433
TCP TTL:103 TOS:0x0 ID:256 IpLen:20 DgmLen:40
*****S* Seq: 0x6D5F0000 Ack: 0x0 Win: 0x4000
TcpLen: 20
[Xref => http://doc.emergingthreats.net/2010935]
```

```
[**] [1:2010935:2] ET POLICY Suspicious inbound
to MSSQL port 1433 [**]
[Classification: Potentially Bad Traffic]
[Priority: 2]
07/25-20:31:31.898603 17.1.27.5:6000 ->
23.19.20.22:1433
TCP TTL:103 TOS:0x0 ID:256 IpLen:20 DgmLen:40
*****S* Seq: 0x262F0000 Ack: 0x0 Win: 0x4000
TcpLen: 20
[Xref => http://doc.emergingthreats.net/2010935]
```

Extraction of relevant patterns and data from Alert files using Notepad++

Once the alert file is generated, it can be opened in Notepad++, which has the unique feature of bookmarking specific lines. To separate the lines that have *Classification*, *Priority* and *IPLen* from the Alert file, the *Find and bookmark* feature can be used.

Once the lines are marked with bookmarks, you can easily cut these selected lines.

Finally, let's paste these lines to a separate file, and the following file will be generated:

```
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen: 233
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen: 233
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen: 234
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen: 232
[Classification: Potentially Bad Traffic]
```

```
[Priority: 2] Dgmlen: 234
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen: 231
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen: 232
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen : 231
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen : 232
[Classification: Potentially Bad Traffic]
[Priority: 2] Dgmlen : 230
```

The extracted alert file can be moved to a spreadsheet package to plot graphs or apply any formula using the following steps:

1. In MS-Word => Use *Find-Replace* for replacing every paragraph with a tab
2. For this, simply replace SPACE with ^t
3. It will generate the file with fields separated by tabs
4. Copy the full file with the tab to MS-Excel
5. MS-Excel will place all tab fields in different columns
6. Now, you can easily apply any formula

Using this method, you can get the following type of output in the Spreadsheet Package:

Classification	Priority	IPLen	dgmLen
Attempted Administrator Privilege Gain	1	20	231
Potentially Bad Traffic	1	20	231
Attempted Administrator Privilege Gain	1	20	231
Potentially Bad Traffic	2	20	231
Attempted Administrator Privilege Gain	1	20	231
Attempted Administrator Privilege Gain	1	20	231
Potentially Bad Traffic	2	20	231
Attempted Administrator Privilege Gain	1	20	23
Attempted Administrator Privilege Gain	1	20	23
Potentially Bad Traffic	2	20	24

This method can be used to analyse any network traffic captured by APIs like PCAP, enabling detailed investigation of the packets and associated parameters. 

By: Dr Gaurav Kumar

The author is associated with various academic and research institutes, where he delivers lectures and conducts technical workshops on the latest technologies and tools. You can contact him at kumargauravin@gmail.com